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Date: 18/03/2026
Duration: 1 hour 45

CONTINUOUS ASSESSMENT 1
II YEAR – SEMESTER 4
CSE23AE204 - PCP III
(B.Tech. E01,E02,E03,E05,E06, E61)

SET F

Question 1: Problem Statement

A list of intervals is provided where each interval represents a start and end time.

Some intervals may overlap with each other.

The task is to **merge all overlapping intervals** and return the resulting set of merged intervals.

Two intervals overlap if:

$\text{current_start} \leq \text{previous_end}$

When overlapping occurs, the merged interval becomes:

$[\text{start_of_first_interval}, \max(\text{end_of_both_intervals})]$

Input Format

`{"intervals": [[start1, end1], [start2, end2], ...]}`

Where:

- `intervals[i][0]` → start time
- `intervals[i][1]` → end time

Output Format

Return the merged intervals:

`[[merged_start1, merged_end1], [merged_start2, merged_end2], ...]`

Example 1

Input

`{"intervals": [[1,3],[2,6],[8,10],[15,18]]}`

Output

`[[1,6],[8,10],[15,18]]`

Explanation

Intervals [1,3] and [2,6] overlap.

Merged interval:

`[1,6]`

Remaining intervals do not overlap.

Final result:

`[[1,6],[8,10],[15,18]]`

Question 2: Problem Statement

You are given a **directed graph** consisting of n nodes numbered from 0 to $n-1$.

The graph is represented by a list of **directed edges**, where each edge is represented as:

`[u, v]`

This means there is a **directed edge from node u to node v** .

Your goal is to find the **minimum set of starting vertices** such that every node in the graph is **reachable from at least one of these starting vertices**.

A node v is considered **reachable** from node u if there exists a path following the directed edges from u to v .

You must return a list of vertices representing the **minimum number of starting nodes** required to reach all nodes in the graph.

If multiple answers are possible, return **any valid list**.

Important notes:

- The graph may contain **multiple disconnected components**.
- The graph is **directed**.

- The returned list should contain the **minimum number of vertices** required to cover all nodes.

Example 1

Input

 $n = 6$

edges = [[0,1],[0,2],[2,5],[3,4],[4,2]]

Output

[0,3]

Explanation

Graph structure:

 $0 \rightarrow 1$ $\downarrow$ $2 \rightarrow 5$ $3 \rightarrow 4 \rightarrow 2$ Nodes **0** and **3** have **no incoming edges**, so they must be included as starting points.

From these two nodes, every other node becomes reachable.

Question 3: Problem StatementYou are given an array of **N non-negative integers**.Your task is to determine the **maximum difference between any two consecutive elements in the sorted version of the array**.However, instead of explicitly sorting the array, the solution should aim to run in **linear time $O(n)$** .If the array contains **fewer than two elements**, return **0**.**Example 1**

Input

4

3 6 9 1

Output

3

Explanation

Sorted array:

[1, 3, 6, 9]

Differences:

 $3-1 = 2$ $6-3 = 3$ $9-6 = 3$ Maximum gap = **3**